

# M. ZARYAB MALIK

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## SUMMARY

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BS Computing, Mathematics & AI student who independently architects and ships production AI systems end-to-end — RAG chatbots, face-recognition pipelines, and campus automation platforms — spanning backend, database, deployment, and Android packaging. Also freelances in graphic design and brand identity.

## TECHNICAL SKILLS

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- Languages: Python, C++, JavaScript, HTML/CSS, SQL
- AI/ML: Retrieval-Augmented Generation (RAG), LLM integration (Groq, Gemini), face recognition (FaceNet, ArcFace, MediaPipe), vector search (ChromaDB, FAISS), prompt engineering
- Backend & Data: FastAPI, PostgreSQL, Firebase, SentenceTransformers
- Deployment: Railway, Netlify, GitHub Pages, PWA Builder (Android APK packaging)
- Tools: Git/GitHub, VS Code, Cursor, Canva, Adobe Photoshop, Adobe Illustrator

## PROJECTS

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### **LGU Chatbot** — *RAG-based University Assistant*

- Architected and deployed a retrieval-augmented generation chatbot end-to-end using FastAPI, ChromaDB, SentenceTransformers, and Groq's LLM API
- Engineered Firebase authentication, persistent chat history, and a CrossEncoder reranker that improved retrieval relevance over baseline vector-similarity search
- Shipped as an installable Android APK via PWA Builder, with backend on Railway and frontend on Netlify

### **LGU Gatepass** — *Smart Campus Exit Management System*

- Independently architected a gate pass management system automating student exit verification through a multi-step HOD approval workflow
- Built the full stack solo — Python/FastAPI backend, PostgreSQL database, face-recognition verification, and an admin dashboard for approve/reject decisions
- Deployed on Railway and Netlify; packaged as an installable Android APK

### **GPS + Face Recognition Attendance System** — *Campus Attendance Automation*

- Engineered a geofenced attendance system combining polygon-based GPS verification with real-time face recognition (FastAPI, ArcFace, FAISS)
- Redesigned the geofence from a circular to a polygon boundary and added a live GPS position indicator on an SVG campus map for accuracy
- Deployed on Railway; packaged as a TWA Android APK

## FREELANCE GRAPHIC DESIGN

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- Delivered product packaging, Amazon listing graphics, and brand identity design for e-commerce clients via Fiverr

## EDUCATION

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### **BS Computing, Mathematics & AI (CMAI)** — Lahore Garrison University, Lahore, Pakistan

2nd Semester | CGPA: 3.39 | Expected Graduation: 2029